第十一章 文件系统实现

1. **What are the advantages of the variation of linked allocation that uses a FAT to chain together the blocks of a file?**

① Much faster sequential and random access.

② Full block is available for data, with no pointer inside blocks

③ Less disk I/O and more caching

④ Easy to block allocation

2. **Some file systems allow disk storage to be allocated at different levels of granularity. For instance, a file system could allocate 4 KB of disk space as a single 4-KB block or as eight 512-byte blocks. How could we take advantage of this flexibility to improve performance? What modifications would have to be made to the free-space management scheme in order to support this feature?**

(1) ① Reduce internal fragmentation for small files.

② Increase efficiency for large sequential access for large files

③ Make the system more flexible.

(2) ① Must track different block sizes.

② Need contiguous groups of small blocks to form a large block.

③ Need to handle splitting and coalescing blocks.

3. **Fragmentation on a storage device could be eliminated by recompaction of the information. Typical disk devices do not have relocation or base registers (such as are used when memory is to be compacted), so how can we relocate files? Give two reasons why recompacting and relocation of files often are avoided.**

(1) Update the file's metadata in inode or FAT pointer

(2) ① Very expensive because of I/O

② Moving files while they are in use may cause durability and reliability problems.

4. **Discuss how performance optimizations for file systems might result in difficulties in maintaining the consistency of the systems in the event of computer crashes.**

Performance optimizations for file systems often delay, batch or reorder writes for efficiency.
After a crash, the file system structure might be broken, data and files might be corrupted.